Rates of Convergence of TD(λ) in Finite Discounted Markov Decision Processes

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Abstract

For the evaluation of a fixed policy in a finite discounted MDP, we study how the trace parameter λ and the choice of features affect two quantities: the quality of the TD(λ) limit (its *bias*) and the speed at which it is reached (its *intrinsic variance*, the asymptotic mean-squared error scaled by the number of samples). Every quantity has a closed form for finite chains, including the long-run noise covariance, which we compute exactly. The results are organized as short statements with proofs; the few results we use without proof are marked as *cited facts*, and patterns supported only by computation are marked as *observations*. The main conclusions are: (i) the limit improves and the signal strengthens as λ increases, with explicit constants; (ii) when the span of the features is invariant under the time-reversed chain, the limit does not depend on λ at all; (iii) the intrinsic variance depends only on the span of the features, and recursive TD is never faster than least-squares TD; (iv) contrary to what the bounds suggest, the intrinsic variance is *not* monotone in λ in general, and can decrease along with the bias. This chapter is the policy-evaluation layer of a study comparing actor-critic and actor-only methods.

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1 Setting

A fixed policy in a finite MDP induces a Markov reward process; we work with that process directly.

Assumption 1.1 (Standing assumptions)**.**

1. The state space $\mathcal{S}$ is finite, P is a row-stochastic matrix on $\mathcal{S}$, and $g : \mathcal{S} \to \mathbb{R}$ is the expected one-stage reward under the policy.
2. P is irreducible and aperiodic, with stationary distribution π; hence $\pi(x) > 0$ for all x. (For a strictly positive policy family this holds for every parameter as soon as it holds for one.)
3. The discount factor satisfies $\gamma \in [0, 1)$, and $\lambda \in [0, 1]$.
4. The feature matrix $\Phi \in \mathbb{R}^{\mathcal{S}\times m}$, with rows $\phi(x)^\top$, has linearly independent columns.
5. Samples are on-policy: (x_k) is the Markov chain with transition matrix P, started from π.

Notation. $D = \operatorname{diag}(\pi)$, $\langle v, w\rangle_D = v^\top D w$, $\|v\|_D^2 = \langle v, v\rangle_D$. $\Pi = \Phi(\Phi^\top D\Phi)^{-1}\Phi^\top D$ is the D-orthogonal projection onto $\operatorname{span}\Phi$. $V = (I - \gamma P)^{-1} g$ is the value function. $\tilde{P}(x, y) = \pi(y)P(y, x)/\pi(x)$ is the time-reversed chain, which is the D-adjoint of P: $\langle Pv, w\rangle_D = \langle v, \tilde{P}w\rangle_D$. We write $c = \gamma\lambda$ and

$$K_\lambda = \gamma(1 - \lambda)\, P\, (I - cP)^{-1}, \qquad \kappa_\lambda = \frac{\gamma(1 - \lambda)}{1 - \gamma\lambda} \in [0, \gamma].$$

Note that $\kappa_0 = \gamma$, $\kappa_1 = 0$, and κ_λ decreases in λ.

The algorithms. With eligibility trace $z_k = c\, z_{k-1} + \phi(x_k)$ and temporal difference $d_k(r) = g(x_k) + \gamma\phi(x_{k+1})^\top r - \phi(x_k)^\top r$:

- *recursive TD(λ)*: $r_{k+1} = r_k + \alpha_k\, d_k(r_k)\, z_k$;
- *LSTD(λ)*: $\hat{r}_n$ solves $\hat{A}_n\hat{r}_n + \hat{b}_n = 0$, where $\hat{A}_n = \frac{1}{n}\sum_{k<n} z_k(\gamma\phi(x_{k+1}) - \phi(x_k))^\top$ and $\hat{b}_n = \frac{1}{n}\sum_{k<n} z_k\, g(x_k)$.

2 The limit and its quality

Both algorithms target the solution of $A_\lambda r + b_\lambda = 0$, where

$$A_\lambda = \Phi^\top D(I - cP)^{-1}(\gamma P - I)\Phi, \qquad b_\lambda = \Phi^\top D(I - cP)^{-1} g.$$

Lemma 2.1. *$\|Pv\|_D \le \|v\|_D$ for all v.*

Proof. By Jensen's inequality and stationarity, $\|Pv\|_D^2 = \sum_x \pi(x)\big(\sum_y P(x, y)v(y)\big)^2 \le \sum_x \pi(x)\sum_y P(x, y)v(y)^2 = \sum_y \pi(y)v(y)^2$. □

Lemma 2.2. *$(I - cP)^{-1}(\gamma P - I) = K_\lambda - I$, and $\|K_\lambda v\|_D \le \kappa_\lambda\|v\|_D$.*

Proof. Write $\gamma P - I = \gamma(1 - \lambda)P - (I - cP)$ and multiply by $(I - cP)^{-1}$, which commutes with P. For the bound, expand $K_\lambda = \gamma(1 - \lambda)\sum_{j\ge 0} c^j P^{j+1}$ and use Lemma 2.1: $\|K_\lambda v\|_D \le \gamma(1 - \lambda)\sum_j c^j\|v\|_D = \kappa_\lambda\|v\|_D$. □

Proposition 2.3 (Signal strength)**.** *For all* $r \in \mathbb{R}^m$*,*

$$r^\top A_\lambda r \le -(1-\kappa_\lambda)\,\|\Phi r\|_D^2, \qquad 1-\kappa_\lambda = \frac{1-\gamma}{1-\gamma\lambda}.$$

In particular A_λ *is invertible and* $r_\lambda^* = -A_\lambda^{-1} b_\lambda$ *is the unique limit. The constant* $1-\kappa_\lambda$ *increases from* $1-\gamma$ *at* $\lambda = 0$ *to* 1 *at* $\lambda = 1$*.*

Proof. By Lemma 2.2, $r^\top A_\lambda r = \langle \Phi r, (K_\lambda - I)\Phi r\rangle_D \le \kappa_\lambda \|\Phi r\|_D^2 - \|\Phi r\|_D^2$ by Cauchy–Schwarz. Since Φ has independent columns, $\|\Phi r\|_D > 0$ for $r \neq 0$. □

Proposition 2.4 (Projected Bellman equation)**.** *Let* $T_\lambda v = (I - cP)^{-1} g + K_\lambda v$*. Then* $V = T_\lambda V$*, and* Φr_λ^* *is the unique fixed point in* $\operatorname{span}\Phi$ *of* ΠT_λ*.*

Proof. $(I - cP)V = g + \gamma PV - cPV = g + \gamma(1-\lambda)PV$, so $V = (I-cP)^{-1}g + K_\lambda V$. For the second claim, $A_\lambda r + b_\lambda = \Phi^\top D(T_\lambda \Phi r - \Phi r)$ by Lemma 2.2, and $\Phi^\top D w = 0$ means $\Pi w = 0$. So $A_\lambda r + b_\lambda = 0$ if and only if $\Pi T_\lambda \Phi r = \Phi r$. Uniqueness: ΠT_λ is a κ_λ-contraction in $\|\cdot\|_D$, as Π is nonexpansive. □

Theorem 2.5 (Quality of the limit)**.**

$$\|\Phi r_\lambda^* - V\|_D \le \frac{\|\Pi V - V\|_D}{\sqrt{1-\kappa_\lambda^2}}.$$

At $\lambda = 1$ *the limit is the best approximation:* $\Phi r_1^* = \Pi V$*.*

Proof. Since $\Phi r_\lambda^* - \Pi V \in \operatorname{span}\Phi$ is D-orthogonal to $\Pi V - V$,

$$\|\Phi r_\lambda^* - V\|_D^2 = \|\Phi r_\lambda^* - \Pi V\|_D^2 + \|\Pi V - V\|_D^2.$$

By Proposition 2.4, $\Phi r_\lambda^* - \Pi V = \Pi T_\lambda \Phi r_\lambda^* - \Pi T_\lambda V = \Pi K_\lambda(\Phi r_\lambda^* - V)$, whose norm is at most $\kappa_\lambda \|\Phi r_\lambda^* - V\|_D$. Rearranging gives the bound; $\kappa_1 = 0$ gives the last claim. □

Proposition 2.6 (Invariant subspaces remove the λ-dependence)**.** *If* $\tilde{P}(\operatorname{span}\Phi) \subseteq \operatorname{span}\Phi$*, then* $\Phi r_\lambda^* = \Pi V$ *for every* $\lambda \in [0,1]$*.*

Proof. The D-adjoint of K_λ is $K_\lambda^* = \gamma(1-\lambda)\sum_j c^j \tilde{P}^{j+1}$, which maps $\operatorname{span}\Phi$ into itself. Hence for $u \in \operatorname{span}\Phi$ and w D-orthogonal to $\operatorname{span}\Phi$, $\langle u, K_\lambda w\rangle_D = \langle K_\lambda^* u, w\rangle_D = 0$, i.e. $\Pi K_\lambda(I - \Pi) = 0$. Then $\Pi T_\lambda(\Pi V) = \Pi T_\lambda V - \Pi K_\lambda(I-\Pi)V = \Pi V$, using $V = T_\lambda V$. By uniqueness (Proposition 2.4), $\Phi r_\lambda^* = \Pi V$. □

Remark 2.7. The hypothesis holds when $\operatorname{span}\Phi$ is spanned by eigenvectors of $\tilde{P}$, for example Fourier features on a random walk on a cycle (circulant P), which is example (b) below. Then the bias–variance trade-off in λ disappears and only the variance matters.

3 Rates: intrinsic variance

Let $(x_k)_{k\in\mathbb{Z}}$ be the stationary two-sided chain, $z_k = \sum_{i\geq 0} c^i \phi(x_{k-i})$ the stationary trace (the series converges because $c = \gamma\lambda \leq \gamma < 1$), and

$$f_k = d_k(r^*_\lambda)\, z_k$$

the noise at the limit. A trace started at time 0 differs from z_k by $c^k z_{-1}$, which does not affect any of the asymptotic statements below.

Cited Fact 3.1 (Markov chain CLT)**.** Under Assumption 1.1, $\mathbb{E}f_k = 0$ (Lemma 4.2), the series $\Gamma = \sum_{l\in\mathbb{Z}} \mathbb{E}[f_0 f_l^\top]$ converges absolutely, and $\frac{1}{\sqrt{n}}\sum_{k<n} f_k$ converges in distribution to $N(0,\Gamma)$. Moreover $\hat{A}_n \to A_\lambda$ almost surely. (Konda 2002, Thm. 6.1, whose conditions are verified for TD in Tsitsiklis and Van Roy 1997; see also Meyn and Tweedie, Ch. 17. Here f_k is a bounded function of the past of a finite, irreducible, aperiodic chain, and its dependence on the distant past decays geometrically because the trace forgets at rate c.)

Theorem 3.2 (Rate of LSTD(λ))**.** *$\sqrt{n}(\hat{r}_n - r^*_\lambda)$ converges in distribution to $N(0,\Sigma_0)$ with $\Sigma_0 = A_\lambda^{-1}\Gamma A_\lambda^{-\top}$. Consequently $n\|\Phi(\hat{r}_n - r^*_\lambda)\|_D^2$ converges in distribution to a random variable with mean equal to the* intrinsic variance

$$\mathrm{IV}(\lambda) = \mathrm{tr}\big(\Sigma_0\, \Phi^\top D \Phi\big).$$

Proof. Since $z_k(g(x_k) + \gamma\phi(x_{k+1})^\top r^* - \phi(x_k)^\top r^*) = f_k$, we have $\hat{b}_n + \hat{A}_n r^* = \frac{1}{n}\sum_{k<n} f_k$, hence exactly

$$\hat{r}_n - r^* = -\hat{A}_n^{-1}\Big(\frac{1}{n}\sum_{k<n} f_k\Big).$$

Multiply by $\sqrt{n}$; by Fact 3.1 and Slutsky's lemma the limit is $-A^{-1}N(0,\Gamma) = N(0, A^{-1}\Gamma A^{-\top})$. By the continuous mapping theorem $n\|\Phi(\hat{r}_n - r^*)\|_D^2 \to \|\Phi w\|_D^2$ in distribution with $w \sim N(0,\Sigma_0)$, and $\mathbb{E}\|\Phi w\|_D^2 = \mathrm{tr}(\Sigma_0\Phi^\top D\Phi)$. □

Remark 3.3. We use the mean of the limiting distribution rather than $\lim_n n\,\mathbb{E}\|\Phi(\hat{r}_n - r^*)\|_D^2$, which can be infinite for finite n when $\hat{A}_n$ is singular with positive probability. In simulations the two agree.

Theorem 3.4 (Only the span of the features matters)**.** *Replace Φ by ΦB for an invertible $B \in \mathbb{R}^{m\times m}$. Then r^*, A, Γ, Σ_0 become $B^{-1}r^*$, $B^\top AB$, $B^\top\Gamma B$, $B^{-1}\Sigma_0 B^{-\top}$. The LSTD approximations $\Phi\hat{r}_n$ are unchanged sample path by sample path, and so are Φr^*_λ and* $\mathrm{IV}(\lambda)$.

Proof. Traces and features transform as $z \mapsto B^\top z$, $\phi \mapsto B^\top\phi$, so $\hat{A}_n \mapsto B^\top\hat{A}_n B$ and $\hat{b}_n \mapsto B^\top\hat{b}_n$, and the solution of the new system is $B^{-1}\hat{r}_n$; thus $\Phi B B^{-1}\hat{r}_n = \Phi\hat{r}_n$. The same computation applies to A, b, r^*. The noise becomes $B^\top f_k$, so $\Gamma \mapsto B^\top\Gamma B$, and $\Sigma_0 \mapsto B^{-1}A^{-1}B^{-\top}B^\top\Gamma B B^{-1}A^{-\top}B^{-\top}$. Finally $\mathrm{tr}(B^{-1}\Sigma_0 B^{-\top}B^\top\Phi^\top D\Phi B) = \mathrm{tr}(\Sigma_0\Phi^\top D\Phi)$. □

3.1 Recursive TD is never faster than LSTD

Cited Fact 3.5 (Rate of recursive TD)**.** With steps $\alpha_k = \alpha/k$ and $K = \alpha A_\lambda + \frac{1}{2}I$ Hurwitz, $\sqrt{k}(r_k - r^*)$ is asymptotically $N(0,\Sigma_{\mathrm{TD}})$, where $K\Sigma_{\mathrm{TD}} + \Sigma_{\mathrm{TD}}K^\top + \alpha^2\Gamma = 0$. (Konda 2002, Thm. 6.1; Benveniste, Métivier and Priouret, Thm. 13, p. 332.)

Lemma 3.6 (Lyapunov equations)**.** *If K is Hurwitz and $KX + XK^\top = -Q$ with $Q \succeq 0$, then $X = \int_0^\infty e^{Kt} Q e^{K^\top t}\, dt \succeq 0$.*

Proof. $\frac{d}{dt}(e^{Kt} X e^{K^\top t}) = e^{Kt}(KX + XK^\top)e^{K^\top t} = -e^{Kt} Q e^{K^\top t}$. Integrate over $[0, \infty)$ and use $e^{Kt} \to 0$. □

Theorem 3.7 (TD versus LSTD)**.** *Under the conditions of Fact 3.5,*

$$\Sigma_{\mathrm{TD}} - \Sigma_0 \succeq 0,$$

with equality if and only if $(\alpha I + A_\lambda^{-1})\Gamma = 0$; when $\Gamma \succ 0$, if and only if $A_\lambda = -\alpha^{-1} I$.

Proof. Write $A = A_\lambda$ and $X = \Sigma_{\mathrm{TD}} - \Sigma_0$. From $\Sigma_0 = A^{-1}\Gamma A^{-\top}$ we get $K\Sigma_0 + \Sigma_0 K^\top = \alpha(\Gamma A^{-\top} + A^{-1}\Gamma) + \Sigma_0$. Hence

$$KX + XK^\top = -\alpha^2\Gamma - \alpha\Gamma A^{-\top} - \alpha A^{-1}\Gamma - A^{-1}\Gamma A^{-\top} = -(\alpha I + A^{-1})\,\Gamma\,(\alpha I + A^{-1})^\top,$$

and Lemma 3.6 gives $X \succeq 0$. The integrand vanishes identically iff $(\alpha I + A^{-1})\Gamma(\alpha I + A^{-1})^\top = 0$, iff $(\alpha I + A^{-1})\Gamma^{1/2} = 0$. □

Remark 3.8. Equality needs A_λ to be a negative multiple of the identity, hence symmetric. For $\lambda = 1$, $A_1 = -\Phi^\top D \Phi$ is symmetric, so TD(1) with $\alpha = 1$ and D-orthonormal features is as fast as LSTD(1). For $\lambda < 1$, A_λ is generally not symmetric, and since a change of basis maps A to $B^\top A B$, which preserves (non)symmetry, recursive TD(λ) is then, when $\Gamma \succ 0$, strictly slower than LSTD(λ) in every basis and for every α. (Polyak–Ruppert averaging of the recursive iterates recovers Σ_0; Konda 2002, Sec. 6.2.1.)

4 Exact computation of the noise covariance

We now give Γ in closed form. Let $r = r_\lambda^*$ and define

$$\begin{aligned} d(x,y) &= g(x) + \gamma\phi(y)^\top r - \phi(x)^\top r, & \bar{d}(x) &= \textstyle\sum_y P(x,y) d(x,y), \\ u &= (I - cP)^{-1}\bar{d} \in \mathbb{R}^{\mathcal{S}}, & W(x) &= \phi(x)\, u(x) \in \mathbb{R}^m, \\ \mu &= (I - c\tilde{P})^{-1}\Phi, & \mu^-(x) &= \textstyle\sum_y \tilde{P}(x,y)\mu(y), \end{aligned}$$

and let $q = \sum_{n \geq 1} P^n W = (Z - I)W$ with $Z = (I - P + \mathbf{1}\pi^\top)^{-1}$ (convergence is shown in Lemma 4.2). Let $M : \mathcal{S} \to \mathbb{R}^{m \times m}$ be the solution of the linear system

$$M(x) = \phi(x)\phi(x)^\top + c(\phi(x)\mu^-(x)^\top + \mu^-(x)\phi(x)^\top) + c^2 \sum_y \tilde{P}(x,y) M(y),$$

which is unique since $c^2 < 1$.

Lemma 4.1 (Moments of the stationary trace)**.** $\mathbb{E}[z_0 \mid x_0] = \mu(x_0)$, $\mathbb{E}[z_{-1} \mid x_0] = \mu^-(x_0)$ *and* $\mathbb{E}[z_0 z_0^\top \mid x_0] = M(x_0)$. *Moreover, given x_0, the trace z_0 is independent of $(x_1, x_2, \dots)$.*

Proof. Run backwards in time, $(x_0, x_{-1}, x_{-2}, \dots)$ is a Markov chain with transition matrix $\tilde{P}$, so $\mathbb{E}[\phi(x_{-i}) \mid x_0] = (\tilde{P}^i\Phi)(x_0)$ and $\mathbb{E}[z_0 \mid x_0] = \sum_i c^i(\tilde{P}^i\Phi)(x_0) = \mu(x_0)$; then $\mathbb{E}[z_{-1} \mid x_0] = \mathbb{E}[\mathbb{E}[z_{-1} \mid x_{-1}] \mid x_0] = (\tilde{P}\mu)(x_0)$. For the second moment, $z_0 = \phi(x_0) + c z_{-1}$, so $\mathbb{E}[z_0 z_0^\top \mid x_0]$ equals $\phi\phi^\top + c(\phi\,\mu^{-\top} + \mu^-\phi^\top) + c^2\mathbb{E}[z_{-1}z_{-1}^\top \mid x_0]$, and by the backward Markov property the last term is $\sum_y \tilde{P}(x_0, y)\mathbb{E}[z_{-1}z_{-1}^\top \mid x_{-1} = y]$. Stationarity makes the conditional second moment the same function M at every time. The independence claim is the Markov property: z_0 is a function of the past. □

Lemma 4.2 (Mean of the noise)**.** $\mathbb{E}f_0 = \Phi^\top Du = \pi^\top W$*, and this equals* $A_\lambda r + b_\lambda = 0$*. Consequently* $\|P^n W\| \to 0$ *geometrically and* q *is well defined.*

Proof. $\mathbb{E}f_0 = \sum_i c^i \mathbb{E}[\phi(x_{-i})\bar{d}(x_0)] = \sum_i c^i \Phi^\top D P^i \bar{d} = \Phi^\top Du$, using $\mathbb{E}[\phi(x_{-i})h(x_0)] = \Phi^\top D P^i h$. Since $\bar{d} = g + (\gamma P - I)\Phi r$, this is $b_\lambda + A_\lambda r = 0$. By aperiodicity and irreducibility $P^n \to \mathbf{1}\pi^\top$ geometrically, so $P^n W \to \mathbf{1}\,\pi^\top W = 0$ geometrically; then $\sum_{n\geq 0} P^n W = \sum_{n\geq 0}(P - \mathbf{1}\pi^\top)^n W = ZW$. □

Lemma 4.3 (Accumulated future noise)**.** *Let* $\mathcal{F}_1 = \sigma(x_j : j \leq 1)$*. As* $L \to \infty$*,*

$$\sum_{l=1}^{L} \mathbb{E}[f_l \mid \mathcal{F}_1] \longrightarrow u(x_1)\, z_1 + q(x_1) - \beta$$

boundedly and almost surely, for a deterministic vector $\beta \in \mathbb{R}^m$*.*

Proof. For $l \geq 1$, $\mathbb{E}[f_l \mid \mathcal{F}_l] = \bar{d}(x_l) z_l$, and $z_l = c^{l-1} z_1 + \sum_{j=2}^{l} c^{l-j}\phi(x_j)$. Taking $\mathbb{E}[\cdot \mid \mathcal{F}_1]$ and summing,

$$\sum_{l=1}^{L} \mathbb{E}[f_l \mid \mathcal{F}_1] = \Big(\sum_{l=1}^{L} c^{l-1}(P^{l-1}\bar{d})(x_1)\Big) z_1 + \sum_{j=2}^{L} \big(P^{j-1}[\Phi \odot u_{L-j}]\big)(x_1),$$

where $u_N = \sum_{i=0}^{N} c^i P^i \bar{d}$ and $(\Phi \odot v)(x) = \phi(x)v(x)$. The first term tends to $u(x_1)z_1$. Write $u_N = u - t_N$ with $\|t_N\|_\infty \leq C c^N$. The part with u tends to $q(x_1)$ by Lemma 4.2. For the tail part, write $P^{j-1} = \mathbf{1}\pi^\top + (P^{j-1} - \mathbf{1}\pi^\top)$ with $\|P^{j-1} - \mathbf{1}\pi^\top\|_\infty \leq C\rho^{j-1}$ for some $\rho < 1$, and set $N = L - j$:

$$\sum_{j=2}^{L} P^{j-1}[\Phi \odot t_{L-j}] = \mathbf{1} \sum_{N=0}^{L-2} \pi^\top[\Phi \odot t_N] + O\Big(\sum_{N=0}^{L-2} \rho^{L-N-1} c^N\Big).$$

The first sum converges absolutely to some β, and the error term tends to zero. All quantities are bounded uniformly in L because Φ, $\bar{d}$ and z are bounded. □

Theorem 4.4 (Closed form for Γ)**.** $\Gamma = E_0 + T + T^\top$*, where*

$$E_0 = \sum_{x,y} \pi(x) P(x,y)\, d(x,y)^2\, M(x),$$

$$T = \sum_{x,y} \pi(x) P(x,y)\, d(x,y) \Big[u(y)\big(c\, M(x) + \mu(x)\phi(y)^\top\big) + \mu(x)\, q(y)^\top\Big].$$

Proof. $E_0 = \mathbb{E}[f_0 f_0^\top] = \mathbb{E}[d(x_0,x_1)^2\, \mathbb{E}[z_0 z_0^\top \mid x_0, x_1]]$, and by Lemma 4.1 the inner expectation is $M(x_0)$. For the off-diagonal part, f_0 is $\mathcal{F}_1$-measurable, so by Lemma 4.3 and bounded convergence,

$$T = \sum_{l\geq 1} \mathbb{E}[f_0 f_l^\top] = \lim_L \mathbb{E}\Big[f_0 \Big(\sum_{l=1}^{L} \mathbb{E}[f_l \mid \mathcal{F}_1]\Big)^\top\Big] = \mathbb{E}[f_0 (u(x_1)z_1 + q(x_1))^\top] - \mathbb{E}[f_0]\beta^\top,$$

and the last term vanishes by Lemma 4.2. Substitute $z_1 = c z_0 + \phi(x_1)$ and use Lemma 4.1: $\mathbb{E}[z_0 z_1^\top \mid x_0, x_1] = c M(x_0) + \mu(x_0)\phi(x_1)^\top$ and $\mathbb{E}[z_0 \mid x_0, x_1] = \mu(x_0)$. Finally, $\sum_{l\leq -1} \mathbb{E}[f_0 f_l^\top] = T^\top$ by stationarity. □

Remark 4.5 (Numerical check). Theorem 4.4 is implemented in `experiments/td_lambda.py`. On the two test chains below, Monte Carlo estimates of $n\,\mathbb{E}\|\Phi(\hat{r}_n - r^*)\|_D^2$ (4000 runs of length 4000) agree with $\mathrm{IV}(\lambda)$ to within 2%, the Monte Carlo error, for $\lambda \in \{0, 0.5, 0.9\}$; and IV is unchanged to six digits under random changes of basis, as Theorem 3.4 predicts.

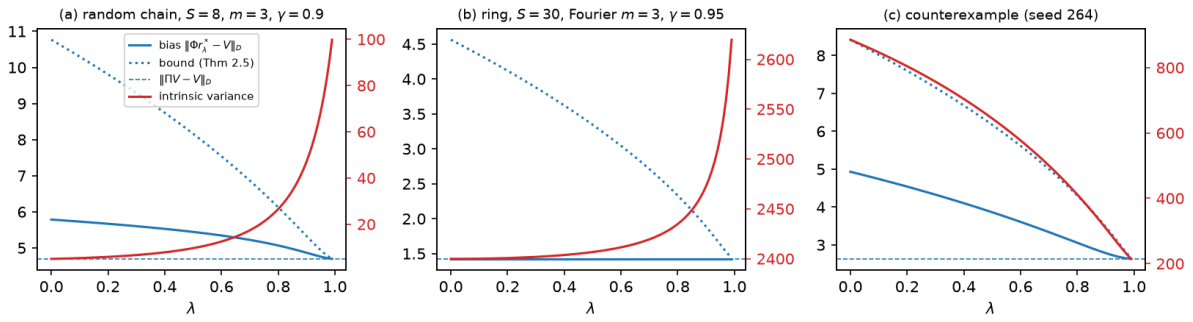


Figure 1: Bias (blue, left axis), the bound of Theorem 2.5 (dotted), the best possible error $\|\Pi V - V\|_D$ (dashed), and intrinsic variance (red, right axis) as functions of λ. (a) a random chain; (b) a slowly mixing ring with Fourier features, where Proposition 2.6 applies and the bias is flat; (c) a chain where the intrinsic variance *decreases* with λ.

5 The trade-off in λ

Definition 5.1. The *asymptotic error proxy* after n samples is $\mathcal{E}_n(\lambda) = \|\Phi r_\lambda^* - V\|_D^2 + \mathrm{IV}(\lambda)/n$, the leading terms of $\mathbb{E}\|\Phi \hat{r}_n - V\|_D^2$; and $\lambda^*(n)$ denotes a minimizer.

The bounds of Section 2 and Konda (2002, Sec. 6.2.2) suggest the classical picture: the bias decreases in λ while the variance increases, so $\lambda^*(n)$ increases from small values towards 1 as n grows. The computations show that the bias half of this picture is robust, but the variance half is not.

Observation 5.2 (Typical behaviour)**.** For the random chain of Figure 1(a) ($|\mathcal{S}| = 8$, $m = 3$, $\gamma = 0.9$), the intrinsic variance grows twentyfold, from 5.0 at $\lambda = 0$ to 99.8 at $\lambda = 0.99$, while the bias falls from 5.78 to 4.70 (the best possible is 4.69). Recursive TD with the best constant α has about twice the intrinsic variance of LSTD at every λ (e.g. 19.4 versus 9.6 at $\lambda = 0.5$).

Observation 5.3 (Frequency of the patterns)**.** Over 300 random chains ($5 \le |\mathcal{S}| \le 14$, $1 \le m \le 4$, $\gamma \in \{0.8, 0.9, 0.95\}$, λ on a grid of 34 values in $[0, 0.99]$):

- TD $\ge$ LSTD held in all 300 cases, as Theorem 3.7 requires; the ratio of intrinsic variances at $\lambda = 0.3$ had median 1.26 and maximum 17.2.
- $\mathrm{IV}(\lambda)$ was monotone increasing in 269 cases and *not* monotone in 31; in one case (Figure 1(c)) it decreased by 76% from its maximum to $\lambda = 0.99$, so that $\lambda \to 1$ is best for every n.
- The bias was monotone decreasing in 269 cases; in the other 31 the largest increase along λ was at most 1.2% of the bias at $\lambda = 0$ (median 0.05%).
- In 281 cases, $\lambda^*(n)$ lay strictly inside $(0, 1)$ for some $n \in \{10, 10^2, 10^3, 10^4\}$, so the trade-off is genuine in most instances.

Observation 5.4 (Features)**.** On the ring of Figure 1(b) with nested Fourier bases of size $m = 1, 3, 5, 7, 9, 15$ and $\lambda = 0.5$, the bias decreases $(10.58, 1.42, 0.40, 0.39, 0.37, 0.19)$, but the intrinsic variance is not monotone: 7065 for $m = 1$, then $2407, 2618, 2838, 2947, 3033$. Enlarging the span can thus lower both bias and variance, or trade one for the other.

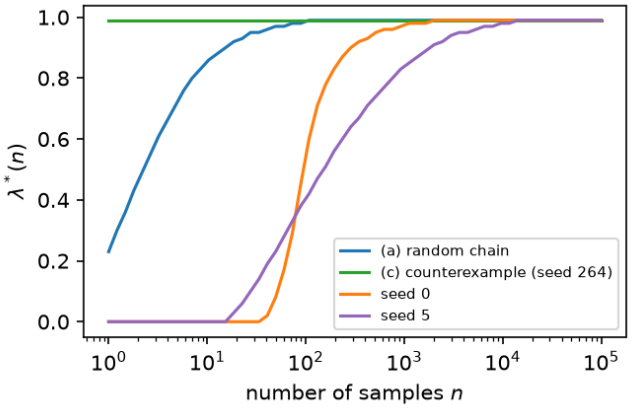


Figure 2: The best trace parameter $\lambda^*(n)$ as a function of the number of samples, for four chains. It moves from small values to 1 as n grows, at a rate set by the ratio of the bias gap to the variance gap; for the chain of Figure 1(c) it equals 1 throughout.

6 Summary and next step

- *Proved.* Larger λ strengthens the signal (Proposition 2.3) and improves the guaranteed quality of the limit (Theorem 2.5); with invariant feature spans, λ does not affect the limit at all (Proposition 2.6). The rate depends only on the span of the features (Theorem 3.4), recursive TD is never faster than LSTD, and for $\lambda < 1$ it is strictly slower in every basis (Theorem 3.7). The long-run noise covariance has a closed form (Theorem 4.4).
- *Observed.* The intrinsic variance usually increases with λ, but not always, and can fall together with the bias; the number of samples at which large λ pays off varies widely between instances.

The next chapter uses the same machinery for the actor: the asymptotic covariance of the actor's parameter in a two-time-scale actor-critic algorithm reduces to a single-time-scale iteration (Konda 2002, Thm. 7.8), whose noise combines the actor's own noise with the critic's noise f_k studied here. That reduction lets us compare, with exact numbers, actor-critic algorithms using TD(λ) critics for $\lambda < 1$ with actor-only methods, the case left open in Section 8.4 of the thesis.

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